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BE (8th Semester)

Examination, April-May, 2014

Branch : IT

DESIGN OF UNIX OPERATING SYSTEM

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Part (a) is compulsory. Answer any two parts from (b), (c), (d). Part (a) carries 2 marks and parts (b), (c), (d) carry 7 marks.

Q. 1. (a) Fill in the blanks :

(i) Number of positional parameter is represented by .....

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(ii) ..... is used to make the Global variable.

(b) Explain Structure of Unix operating system.

(c) Write commands for the following :

(i) To set the various Read Write and Execute permission for the file.

(ii) To display list of all C files having 4 character long in a directory cprog.h.

(iii) To print all lines of file in which sum of 1st and 2nd field is equal to 3rd field.

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(iv) Replace all occurrences 'a' with '*', 'e' with '#' and 'i' with '%' in a file sample.

(v) Create link of a file.

(vi) Substitute all occurrences of 'UNIX' to 'LINUX' in a given file.

(d) What do you mean by the term 'File System'? Explain the contents of Boot block, Super block, Inode block and Data block.

Q. 2. (a) Write any four system shell variable.

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(b) Explain use of grep Command with its syntax; giving five suitable examples.

(c) Write shell script to display Multiple Answer type question, accept answer from the user and display proper message.

(d) What is shell script ? What are the different ways to execute shell script ?

Q. 3. (a) Explain the concept of Buffering.

(b) Explain structure of Buffers. Name and discuss the algorithm that kernel executes to get the buffer corresponding to any block.

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(c) What are the contents of Buffer header ? Discuss advantages and disadvantage of buffering mechanism.

(d) Discuss kernel's algorithm for Read/Write operation using buffer.

Q. 4. (a) How in-core copy of Inode is different from disk copy ?

(b) What will be physical address corresponding to logical address 8000 in file. Also discuss the structure of regular file.

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(c) How pipes are different from regular file. Explain how inter process communication takes place using pipes.

(d) Explain alloc( ) and free( ) algorithm used by kernel to allocate and free data block from and to the disk.

Q. 5. (a) Explain fork ( ) system call.

(b) In Unix what are the different states of a process ? How transition of a process occurs between these states.

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(c) Explain the use of Sleep ( ) system call.

(d) Explain the concept of region, region table and virtual address space of a process.

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